

SKM 200 MLI 066 T



SEMITRANS® 5

Trench IGBT Modules

SKM 200 MLI 066 T

Target Data

Features

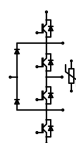
- Homogeneous Si
- Trench = Trenchgate technology
- $V_{CE(sat)}$ with positive temperature coefficient
- Integrated NTC temperature sensor

Typical Applications

- UPS
- 3 Level Inverter

Remarks

- Case temperature limited to $T_c = 125^\circ\text{C}$ max, recommended $T_{op} = -40..+150^\circ\text{C}$



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Absolute Maximum Ratings		T _{case} = 25°C, unless otherwise specified		
Symbol	Conditions		Values	Units
IGBT				
V _{CES}	T _j = 25 °C		600	V
I _C	T _j = 175 °C	T _c = 25 °C	280	A
		T _c = 80 °C	210	A
I _{CRM}	I _{CRM} =2xI _{Cnom}		400	A
V _{GES}			± 20	V
t _{psc}	V _{CC} = 360 V; V _{GE} ≤ 15 V; T _j = 150 °C V _{CES} < 600 V		6	μs
Inverse Diode				
I _F	T _j = 175 °C	T _c = 25 °C	270	A
		T _c = 80 °C	200	A
I _{FRM}	I _{FRM} =2xI _{Fnom}		400	A
I _{FSM}	t _p = 10 ms; half sine wave T _j = 150 °C		1310	A
Freewheeling Diode				
I _F	T _j = 175 °C	T _c = 25 °C	270	A
		T _c = 80 °C	200	A
I _{FRM}	I _{FRM} =2xI _{Fnom}		400	A
I _{FSM}	t _p = 10 ms; half sine wave T _j = 150 °C		1310	A
Module				
I _{t(RMS)}			500	A
T _{vj}			- 40 ... + 175	°C
T _{stg}			- 40 ... + 125	°C
V _{isol}	AC, 1 min.		2500	V

Characteristics		$T_{case} = 25^\circ\text{C}$, unless otherwise specified			
Symbol	Conditions	min.	typ.	max.	Units
IGBT					
$V_{GE(th)}$	$V_{GE} = V_{CE}, I_C = 3,2\text{ mA}$	5	5,8	6,5	V
I_{CES}	$V_{GE} = 0\text{ V}, V_{CE} = V_{CES} \quad T_j = 25^\circ\text{C}$			0,01	mA
I_{GES}	$V_{CE} = 0\text{ V}, V_{GE} = 20\text{ V} \quad T_j = 25^\circ\text{C}$			1200	nA
V_{CE0}			$T_j = 25^\circ\text{C}$	0,9	V
			$T_j = 150^\circ\text{C}$	0,7	V
r_{CE}	$V_{GE} = 15\text{ V}$		$T_j = 25^\circ\text{C}$	2,7	$\text{m}\Omega$
			$T_j = 150^\circ\text{C}$	5	$\text{m}\Omega$
$V_{CE(sat)}$	$I_{Cnom} = 200\text{ A}, V_{GE} = 15\text{ V}$		$T_j = 25^\circ\text{C}_{chiplev.}$	1,45	V
			$T_j = 150^\circ\text{C}_{chiplev.}$	1,7	V
C_{ies}	$V_{CE} = 25, V_{GE} = 0\text{ V} \quad f = 1\text{ MHz}$			12,3	nF
C_{oes}				0,76	nF
C_{res}				0,36	nF
R_{Gint}	$T_j = ^\circ\text{C}$		1		Ω
$t_{d(on)}$	$R_{Gon} = 2,4\ \Omega$				ns
t_r					ns
E_{on}	$R_{Goff} = 2,4\ \Omega$				mJ
$t_{d(off)}$					ns
t_f					ns
E_{off}					mJ
$R_{th(j-c)}$	per IGBT		0,21		K/W



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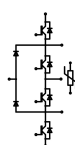
Remarks

- Case temperature limited to $T_C = 125^\circ\text{C}$ max, recommended $T_{op} = -40..+150^\circ\text{C}$

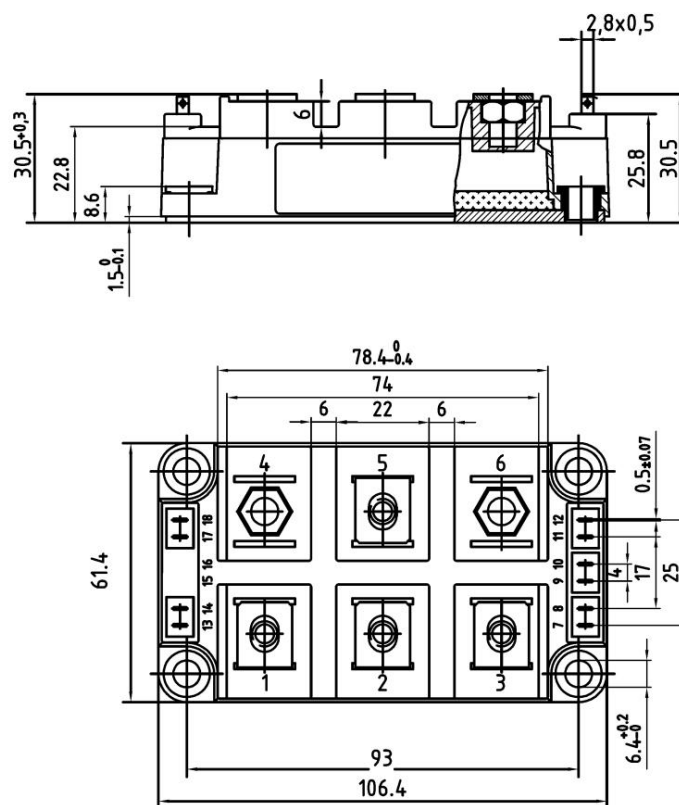
Characteristics					
Symbol	Conditions	min.	typ.	max.	Units
Inverse Diode					
$V_F = V_{EC}$	$I_{Fnom} = 200\text{ A}; V_{GE} = 0\text{ V}$	$T_j = 25\text{ }^{\circ}\text{C}_{chiplev.}$	1,4	1,6	V
		$T_j = 150\text{ }^{\circ}\text{C}_{chiplev.}$	1,4	1,6	V
V_{F0}		$T_j = 25\text{ }^{\circ}\text{C}$	0,95	1	V
		$T_j = 150\text{ }^{\circ}\text{C}$	0,85	0,9	V
r_F		$T_j = 25\text{ }^{\circ}\text{C}$	2	3	mΩ
		$T_j = 150\text{ }^{\circ}\text{C}$	2,7	3,5	mΩ
I_{RRM}	$I_F = 200\text{ A}$	$T_j = 150\text{ }^{\circ}\text{C}$			A
Q_{rr}					μC
E_{rr}	$V_{GE} = -8\text{ V}; V_{CC} = 300\text{ V}$				mJ
$R_{th(j-c)D}$	per diode		0,39		K/W
Free-wheeling diode (Neutral Clamp Diode)					
$V_F = V_{EC}$	$I_{Fnom} = 200\text{ A}; V_{GE} = 0\text{ V}$	$T_j = 25\text{ }^{\circ}\text{C}_{chiplev.}$	1,4	1,6	V
		$T_j = 150\text{ }^{\circ}\text{C}_{chiplev.}$	1,4	1,6	V
V_{F0}		$T_j = 25\text{ }^{\circ}\text{C}$	0,95	1	V
		$T_j = 150\text{ }^{\circ}\text{C}$	0,85	0,9	V
r_F		$T_j = 25\text{ }^{\circ}\text{C}$	2	3	V
		$T_j = 150\text{ }^{\circ}\text{C}$	2,7	3,5	V
I_{RRM}	$I_F = 200\text{ A}$	$T_j = 150\text{ }^{\circ}\text{C}$			A
Q_{rr}					μC
E_{rr}	$V_{GE} = 0\text{ V}; V_{CC} = 600\text{ V}$				mJ
$R_{th(j-c)FD}$	per diode		0,39		K/W
M_s	to heat sink M6	3		5	Nm
M_t	to terminals M6	2,5		5	Nm
w				310	g
Temperature sensor					
R_{100}	$T_s = 100^{\circ}\text{C}$ ($R_{25} = 5\text{k}\Omega$)	493±5%			Ω K

This is an electrostatic discharge sensitive device (ESDS), international standard IEC 60747-1, Chapter IX.

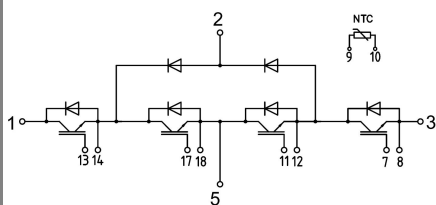
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Case D60



MLI-T

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